

S.W.A.G

Virtual Closet

Software Engineering Project

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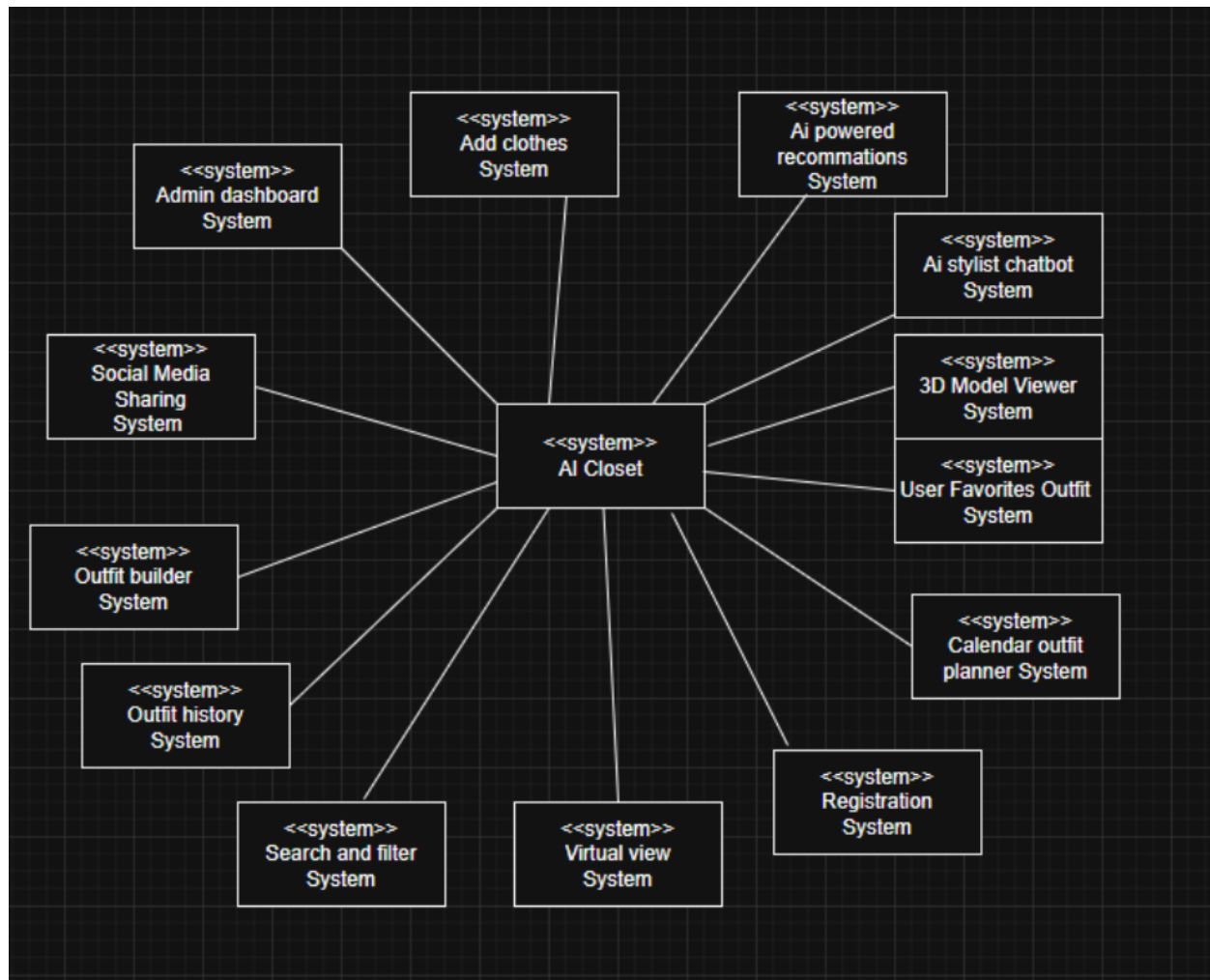
Section 2.1

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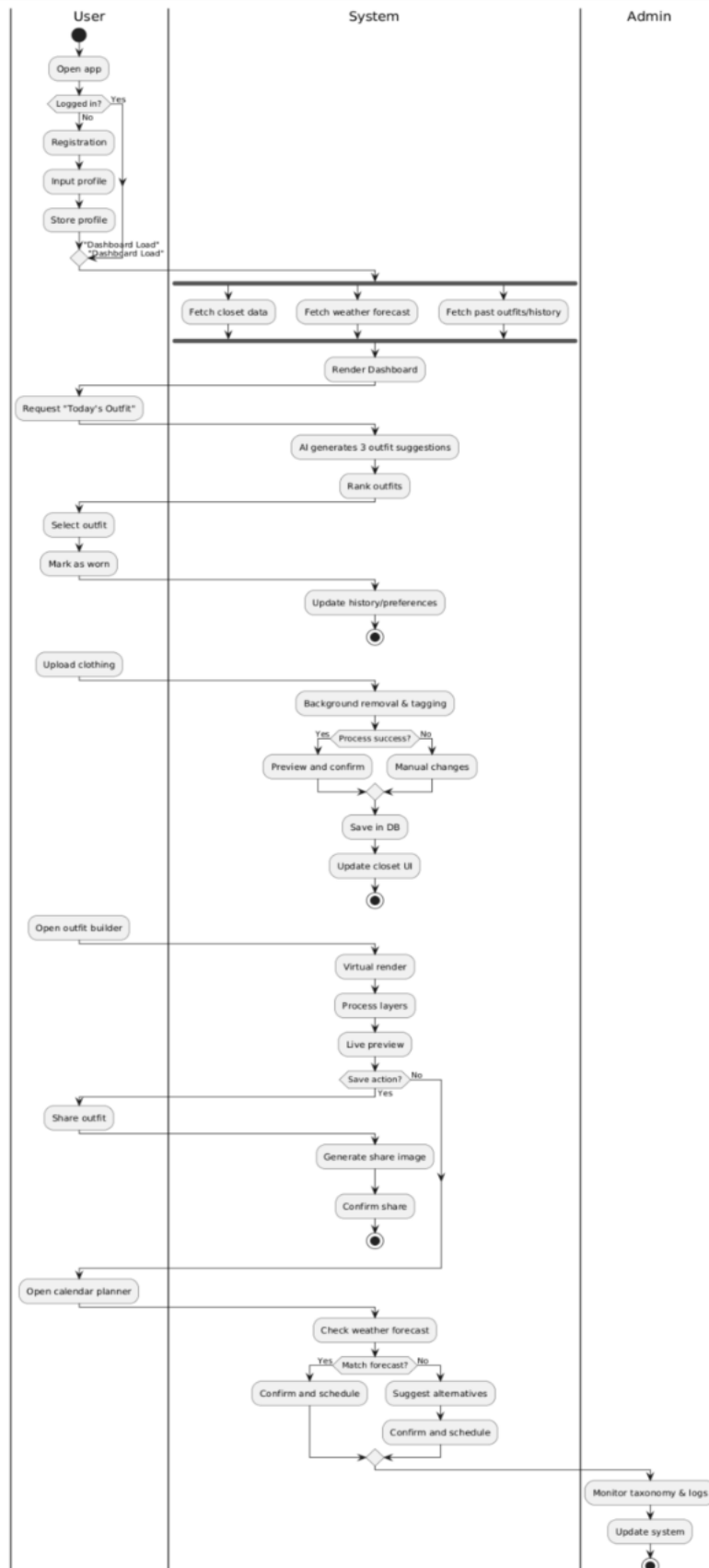
Section 2.2

Problem Statement: S.W.A.G. (Smart Wear AI Guide) is a webapp that acts as a virtual closet and AI stylist, helping users decide what to wear based on their clothes, the weather, and location. Students, professionals, and style-conscious individuals who want quick, weather-ready outfit suggestions. Saves time choosing outfits, prevents repeating looks too often, and ensures clothes match weather conditions. It combines daily weather checks (something users already do) with personal closet management, making it both practical and engaging.

Section 2.3



Section 2.4



Section 2.5

Section 2.5.1

Use Case No: UC1

Use Case Name: User Registration

Actors: User, System, Database

Description: The user opens the registration page and enters personal details such as name, email, and password. The system validates the input fields, checks if the email already exists in the database, and encrypts the password before storing it. Once successfully created, the system confirms registration through a message or verification email.

Alternate Path: User logs in with Google/SSO instead of manual registration.

Exception Path: Invalid or duplicate email, weak password.

Pre-condition: User is not registered.

Post-condition: User account created successfully

Use Case No: UC2

Use Case Name: User Login

Actors: User, System, Database

Description: User navigates to the login page and provides valid credentials. The system verifies the credentials against stored records in the database. If authentication succeeds, the user session begins, and the dashboard loads with access to all wardrobe features.

Alternate Path: Login via SSO.

Exception Path: Wrong password, locked account.

Pre-condition: Users must already be registered.

Post-condition: User logged into system dashboard.

Use Case No: UC3

Use Case Name: Upload Clothing Item

Actors: User, System, AI Image Service, Database

Description: User uploads a clothing image. The system forwards the image to an AI Image Service, which removes the background and tags features such as color, type, and season. The processed image and its metadata are stored in the user's closet database.

Alternate Path: User skips auto-tagging and manually enters clothing details.

Exception Path: Invalid file format, corrupted image, API error.

Pre-condition: User logged in.

Post-condition: Item saved in user's closet.

Use Case No: UC4

Use Case Name: Outfit Builder

Actors: User, System, Database

Description: User selects clothing items from their uploaded closet to build a complete outfit. The system displays chosen items together and saves the outfit configuration in the database.

Alternate Path: User edits outfit (replaces/removes items) before saving.

Exception Path: Selected items not available (e.g., missing tags), database save error.

Pre-condition: User logged in with existing clothing items in closet.

Post-condition: Outfit is created and stored for future reference.

Use Case No: UC5

Use Case Name: Outfit History

Actors: User, System, Database

Description: User views a timeline of previously worn or saved outfits. The system retrieves outfit history from the database and displays them chronologically.

Alternate Path: User filters history by date range, season, or event.

Exception Path: No outfits found, DB query failure.

Pre-condition: User logged in with at least one saved outfit in history.

Post-condition: Previous outfits displayed to user for browsing or reuse.

Use Case No: UC6

Use Case Name: Favorites

Actors: User, System, Database

Description: User marks selected outfits as favorites for quick access. The system flags these outfits in the database and displays them in a separate favorites list.

Alternate Path: User un-favorites an outfit, removing it from the list.

Exception Path: DB error while updating favorites, invalid outfit reference.

Pre-condition: User logged in with at least one outfit saved.

Post-condition: Outfit saved in favorites list for easy retrieval.

Use Case No: UC7

Use Case Name: Request outfit recommendation

Actors: User, System, Weather API, AI Recommendation Engine, Database

Description: The user selects "Today's Outfit." The system calls the Weather API to retrieve current and forecast conditions (temperature, precipitation, humidity). The system then analyzes available clothing items, user preferences, and previous outfits to generate three optimized outfit suggestions using the AI recommendation model. Each recommendation includes images, weather compatibility, and style notes.

Alternate Path: The user filters recommendations (e.g., "Business Casual," "Workout," or "Outdoor").

Exception Path: Weather API unavailable or insufficient closet items.

Pre-condition: User has uploaded clothes and set style preferences.

Post-condition: Personalized outfit recommendations displayed to the user.

Use Case No: UC8

Use Case Name: Select Outfit to wear

Actors: User, System, Database

Description: After reviewing recommendations, the user selects an outfit to wear. The system records this outfit choice in the database, logs the date and weather data, and updates the wear history. This information trains the AI model to refine future suggestions.

Alternate Path: The user can mark a manually created outfit as worn.

Exception Path: Database update failure.

Pre-condition: Recommendations or saved outfits exist.

Post-condition: The outfit is recorded in the user's history

Use Case No.: UC9

Use Case Name: Calendar Outfit Planning

Actors: User, System, Database

Description: User accesses a built-in calendar to schedule outfits for future dates or events. The system allows selecting any saved outfit and linking it to a date on the calendar. Planned outfits can later be edited or deleted.

Alternate Path: Calendar syncs with Google Calendar to send event reminders.

Exception Path: Invalid date or duplicate entry.

Pre-condition: User logged in with at least one saved outfit.

Post-condition: Calendar successfully updated with scheduled outfits.

Use Case No: UC10

Use Case Name: Manage Preferences

Actors: User, System, Database

Description: User customizes personal preferences, including preferred styles, favorite colors, and clothing fit. The system stores this information and applies it as filters in AI recommendation generation.

Alternate Path: User resets preferences to default values.

Exception Path: Invalid input or database failure.

Pre-condition: User logged in.

Post-condition: Updated preferences saved for personalized suggestions.

Use Case No: UC11

Use Case Name: Ai Chatbot Styling suggestions

Actors: User, AI Chatbot, System, Database

Description: User interacts with the built-in AI stylist chatbot via text interface. The chatbot uses NLP to interpret user queries (e.g., "What should I wear to a wedding?"), accesses closet and weather data and provides real-time advice or outfit ideas.

Alternate Path: User asks non-fashion-related questions, and chatbot provides polite default responses.

Exception Path: Chatbot service timeout or processing error.

Pre-condition: User logged in and chatbot service available.

Post-condition: Chatbot provides text-based fashion suggestions and stores the chat session log.

Use Case No: UC12

Use Case Name: Admin Login

Actors: Admin, System, Database

Description: Admin navigates to the admin portal and enters credentials. The system verifies them against admin records and grants access to the admin dashboard with monitoring and management privileges.

Alternate Path: Admin logs in using institutional SSO.

Exception Path: Unauthorized access or inactive admin account.

Pre-condition: Admin credentials exist in the database.

Post-condition: Admin gains access to restricted management tools.

Use Case No: UC13

Use Case Name: Manage Clothing taxonomy

Actors: Admin, System, Database

Description: Admin maintains the master category taxonomy for clothing (e.g., tops, outerwear, formalwear). Updates made here propagate throughout the system for consistent tagging.

Alternate Path: Admin adds seasonal or trend-based tags.

Exception Path: Invalid input or duplicate category name.

Pre-condition: Admin logged in.

Post-condition: Updated taxonomy visible to all users and AI processes.

Use Case No: UC14

Use case name: Monitor System Login

Actors: Admin, System, Database

Description:

Admin accesses system logs to monitor user activity, login attempts, API calls, and errors. Logs can be filtered by date and exported for auditing.

Alternate Path: Logs exported as CSV or PDF.

Exception Path: Log retrieval error or missing permissions.

Pre-condition: Admin logged in.

Post-condition: Logs displayed or downloaded for monitoring.

Use Case No: UC15

Use Case Name: Logout

Actors: User, System

Description: User clicks "Logout." The system ends the active session, clears cached tokens, and redirects the user to the login page.

Alternate Path: Auto-logout occurs after inactivity.

Exception Path: Session not cleared properly due to timeout.

Pre-condition: User logged in.

Post-condition: User securely logged out and redirected to login screen.

Section 2.5.2

Requirement number: R1

Use Case number: UC1

Introduction: System must allow new users to register.

Inputs: Username, Email, Password.

Requirement Description: Functional requirement — system must validate email and password strength.

Outputs: User account created in DB

Requirement number: R2

Use Case number: UC2

Introduction: Users must be able to log in securely.

Inputs: Email, Password.

Requirement Description: Authentication with password encryption; login session management.

Outputs: User successfully logged in.

Requirement number: R3

Use Case number: UC3

Introduction: Users can upload clothing.

Inputs: Image file.

Requirement Description: Background removal, tagging, metadata storage in DB.

Outputs: Clothing saved in user closet

Requirement number: R4

Use Case number: UC4 (Outfit Builder)

Introduction: Users must be able to create complete outfits from items in their closet.

Inputs: Clothing item selections from the user.

Requirement Description: Functional requirement — the system must allow users to select items, display the combined outfit visually, and provide editing options (replace/remove). The system must validate item availability (tags/metadata) before saving.

Outputs: Outfit configuration saved in the database and displayed to the user.

Requirement number: R5

Use Case number: UC5 (Outfit History)

Introduction: Users must be able to view past outfits.

Inputs: User's request for outfit history (with optional filters such as date, season, or event).

Requirement Description: The system must retrieve stored outfits from the database and display them chronologically. Filtering must be supported based on user-specified criteria. System should handle cases where no outfits are found.

Outputs: List of previous outfits shown to the user (filtered or full).

Requirement number: R6

Use Case number: UC6 (Favorites)

Introduction: Users must be able to favorite specific outfits for quick access.

Inputs: User selects an outfit to mark/unmark as favorite.

Requirement Description: the system must update and display outfit favorite status in the database.

Outputs: Favorites list updated and shown to the user.

Requirement number 7

Use Case number: UC7 – Request Outfit Recommendation

Introduction: Users can get outfit suggestions based on weather and preferences.

Inputs: Current weather data, user closet, preferences

Requirement Description: The system connects to the Weather API, checks suitable clothes, and uses AI to recommend 3 outfits.

Outputs: AI-generated outfit suggestions are displayed.

Requirement number 8

Use Case number: UC8 – Select Outfit to Wear

Introduction: Users can select which outfit they are wearing today.

Inputs: Selected outfit ID

Requirement Description: The system marks the chosen outfit as “worn” and stores it in the history with the date.

Outputs: The outfit is saved in the user’s history.

Requirement number 9

Use Case number: UC9 – Calendar Outfit Planning

Introduction: Users can plan future outfits using a calendar.

Inputs: Date, selected outfit

Requirement Description: The system saves the outfit and date together in the database for future reminders.

Outputs: Planned outfits appear on the user’s calendar.

Requirement number 10

Use Case number: UC10 – Manage Preferences

Introduction: Users can set style preferences for recommendations.

Inputs: Favorite colors, preferred styles, clothing fit

Requirement Description: The system stores these preferences and uses them when generating outfit ideas.

Outputs: Updated preferences are saved successfully

Requirement number 11

Use Case number: UC11 – AI Chatbot Styling Suggestions

Introduction: Users can chat with the AI stylist for fashion advice.

Inputs: User's text messages

Requirement Description: The chatbot reads the message, checks closet data, and replies with styling tips or outfit suggestions.

Outputs: The chatbot displays helpful fashion responses.

Requirement number 12

Use Case number: UC12 – Admin Login

Introduction: Admins can log in to manage the system.

Inputs: Admin username and password

Requirement Description: The system checks admin credentials and gives access to the admin dashboard.

Outputs: Admin successfully logs in.

Requirement number 13

Use Case number: UC13 – Manage Clothing Taxonomy

Introduction: Admins can edit or update clothing categories.

Inputs: Category name, type, or tags

Requirement Description: The system lets admins add, edit, or delete clothing categories used by all users.

Outputs: Clothing taxonomy updates across the system.

Requirement number 14

Use Case number: UC14 – Monitor System Logs

Introduction: Admins can view activity and system logs.

Inputs: Date or filter options

Requirement Description: The system retrieves logs of user activities and system events for

review or export.

Outputs: Logs displayed or downloaded for monitoring.

Requirement number 15

Use Case number: UC15 – Logout

Introduction: Users can safely log out of the system.

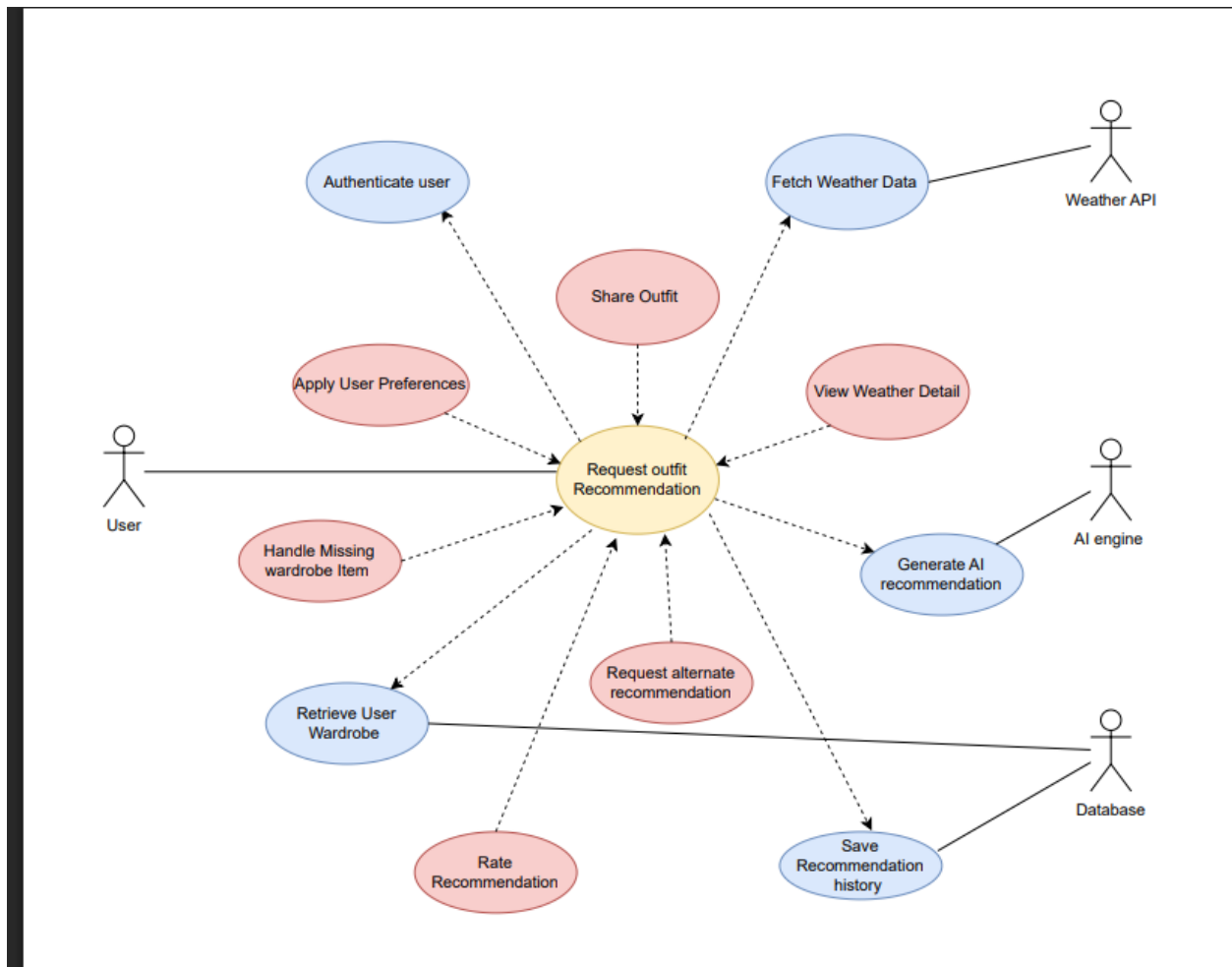
Inputs: Logout command

Requirement Description: The system ends the user session, clears data, and redirects to the login screen.

Outputs: User logged out successfully.

Section 2.5.3

Use case 7 Diagram



Section 2.6

Section 2.7